



ALA-TOO INTERNATIONAL UNIVERSITY

BUSINESS ADMINISTRATION SOFTWARE

Spring Semester 2025-2026

*If you're not paying attention to your customers,
someone else is.*

— Jeff Bezos

CRM MATTERS

CONTEXT



Modern companies do not fail because they cannot produce products. They fail because they lose customers, misunderstand demand, or cannot scale customer relationships. CRM systems exist to solve this problem.

BEFORE CRM

HOW CUSTOMER DATA WAS MANAGED



Before CRM systems, customer information was scattered across emails, spreadsheets, notebooks, personal contacts, and individual employee knowledge. Customer relationships lived in people's heads, not in systems.

PROBLEM WITH THE OLD WAY

WHY IT BREAKS

When customer data is informal (this approach does not scale well):

- Salespeople leave → customer history is LOST
- Customers are contacted inconsistently
- Management CANNOT forecast revenue
- Decisions are based on intuition, not data

REAL-LIFE ANALOGY

CRM AS A MEMORY SYSTEM



CRM is like a shared organizational memory.

Without it, each employee remembers customers differently. But with CRM system the company remembers customers consistently, even when employees change or departments restructure.

CUSTOMER RELATIONSHIP MANAGEMENT

PRECISE DEFINITION



Customer Relationship Management (CRM) is a software system that centrally stores customer data and manages all interactions, sales activities, and communication across the customer lifecycle.

WHAT CRM ACTUALLY STORES

CORE DATA

CRM systems store (this data becomes the basis for decisions):

- Customer profiles (who they are)
- Interaction history (what happened)
- Sales status (what stage they are in)
- Communication records (emails, calls, meetings)

CRM AND BUSINESS VALUE

WHY DATA MATTERS



Customer data is not administrative information.
It directly determines revenue forecasting, marketing effectiveness, sales productivity, and customer retention.

SALES WITHOUT CRM

WHY IT FAILS

Sales without CRM is acceptable for small teams, not for growing organizations. Without CRM:

- Sales pipelines are informal
- Forecasts are guesses
- Managers rely on verbal updates
- Revenue becomes unpredictable

SALES WITH CRM

STRUCTURED PIPELINE



CRM systems enforce sales pipelines with defined stages (lead, opportunity, proposal, closed). Each deal progresses through measurable steps, enabling tracking and forecasting.

SALESFORCE

REAL EXAMPLE



Salesforce was founded specifically to solve the problem of managing customer relationships at scale. Today, it is the world's largest CRM provider.

SALESFORCE AT SCALE

REAL DATA



Salesforce serves 150,000+ organizations globally and generates over \$34 billion in annual revenue, proving the economic importance of structured customer management.

AMAZON

REAL EXAMPLE



(CRM at EXTREME SCALE)

Amazon tracks customer purchases, browsing behavior, reviews, returns, and support interactions. This CRM-driven data enables personalized recommendations that generate 35%+ of Amazon's revenue.

CRM AS A DIGITAL SALES ASSISTANT

ANALOGY

Humans forget. Systems do not. CRM acts like a digital assistant that never forgets:

- Who the customer is
- What they bought
- What problems they had
- What they might need next

CRM VS ERP

CLEAR BOUNDARY

CRM manages customers and revenue generation

CRM answers: Who is buying?

ERP manages operations and execution

ERP answers: How do we deliver?

CRM-ERP INTEGRATION

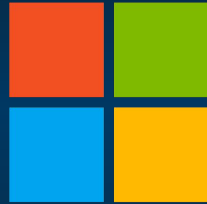
WHY BOTH ARE REQUIRED

Without ERP integration, CRM becomes isolated and incomplete. A sale recorded in CRM must trigger:

- Inventory checks
- Invoicing
- Shipping
- Financial reporting

MICROSOFT

REAL EXAMPLE



Microsoft

Microsoft uses Dynamics CRM integrated with Dynamics ERP to manage enterprise customers, linking sales activity directly to billing, contracts, and service delivery. (Integrated systems)

CRM DATA QUALITY

THE HIDDEN RISK



CRM systems fail NOT because of technology, but because of poor data quality: duplicates, outdated records, missing interactions, and inconsistent usage.

DATA COSTS MONEY

REAL FAILURE EXAMPLE



IBM studies show that poor data quality costs organizations \$3 - 5 trillion annually worldwide, including lost sales, failed campaigns, and bad decisions much of it CRM-related.

CRM AND IT SPECIALISTS

REAL ROLE

This is systems engineering, not UI work. IT specialists:

- Configure CRM workflows
- Integrate CRM with ERP and analytics
- Control access and data integrity
- Ensure reporting accuracy

CRM FOR IT CAREERS

WHY YOU SHOULD CARE

Few systems influence money as directly as CRM. Understanding CRM allows IT professionals to:

- Work with sales and marketing teams
- Support revenue-critical systems
- Build integrations that directly impact profit

CRM PIPELINE DATA

DEFINITION

CRM pipeline data represents how potential customers move through defined sales stages, from first contact to closed deal. Each stage reflects a business decision point and is recorded automatically in the CRM system as sales activities occur. Typical pipeline stages include:

- Lead
- Qualified opportunity
- Proposal
- Negotiation
- Closed (won or lost)

At every stage, the CRM stores structured data such as deal value, probability, owner, and status.

ACTIVITY

CRM PIPELINE CALCULATION



The following activity uses simplified CRM pipeline logic to calculate expected revenue. The goal is to show how structured customer data becomes a concrete financial metric used in real companies.

Use CRM pipeline data to calculate expected revenue and demonstrate how customer information supports measurable business performance.

SCENARIO

PIPELINE DATA

200 leads

Potential customers who have shown initial interest but have not yet been qualified or contacted in depth.

30% convert to opportunities

The proportion of leads that become serious sales opportunities after qualification by the sales team.

20% close rate

The percentage of sales opportunities that result in a successful sale.

Average deal value: \$2,000

The typical revenue generated from one successfully closed deal.

TASK

CLASSWORK

Calculate:

- 1 - Number of closed deals
- 2 - Expected revenue
- 3 - Which CRM data fields are critical

SOLUTION

CLASSWORK

Step 000 : Identify opportunities

Formula: Total leads × conversion rate

Calculation: 200 leads × 30% = 60

Opportunities = 60

SOLUTION

CLASSWORK

Step 001 : Determine closed deals

Formula: Opportunities × close rate

Calculation: 60 opportunities × 20% = 12

Closed deals = 12

SOLUTION

CLASSWORK

Step 002 : Calculate expected revenue

Formula: closed deals × average deal value

Calculation: 12 deals × \$2,000 = \$24,000

Expected revenue = \$24,000

CORRECT ANSWER

VALIDATION

1. Number of closed deals: 12

2. Expected revenue: \$24,000

3. Critical CRM data fields:

Lead status: to track which leads are qualified.

Deal probability: to understand the likelihood of closing at each stage.

Deal value: to calculate the potential financial impact.

Activity history: to ensure consistent communication and prevent loss of data.

WHAT IF

CRM DATA IS WRONG



Imagine a CRM showing strong customer growth, but half the records are duplicates. Management increases marketing spend, expecting growth but real demand does not exist.

REAL CONSEQUENCE

WHY THIS HAPPENS

The system worked but the data did not. Companies using Salesforce and similar platforms frequently report that inaccurate CRM data leads to:

- Incorrect sales forecasts
- Wasted marketing budgets
- Loss of customer trust

CRM MAINTENANCE

ONGOING RESPONSIBILITY

CRM is not “set and forget” it requires continuous effort:

- Data governance
- User discipline
- Integration monitoring
- Workflow refinement

FINAL TAKEAWAY

WHAT CRM REALLY IS



CRM systems transform customer relationships from personal memory into organizational intelligence. Without CRM, customer management does not scale. With CRM, revenue becomes measurable and manageable.

THE LEAKY FUNNEL (REVENUE PREDICTOR)

PRACTICAL P000

You have been hired as a developer for "AlaToo industries", the sales manager is tired of doing manual calculations in Excel. They need create a script that can predict monthly revenue based on different "Lead quality" scenarios.

The company is worried about a "Leaky Funnel" where leads drop out of the system because they aren't contacted fast enough.

THE GOAL

PRACTICAL P000

Write a program (Python or Java) that:

000 - Asks the user for the number of initial Leads.

001 - Calculates how many become Opportunities (30% conversion).

002 - Calculates how many Close (20% close rate).

003 - If the number of leads is over 500, the system becomes "overwhelmed," and the close rate drops from 20% to 10% due to poor customer service.

004 - Outputs the Total Expected Revenue (using \$2,000 per deal).

EXPECTED OUTPUT

PRACTICAL P000

```
=====
  ALA-TOO CRM: REVENUE PREDICTOR
=====
Enter number of incoming leads: 200

[STATUS]: Sales team operating at normal capacity.
-----
PIPELINE ANALYSIS:
> Qualified Opportunities: 60
> Estimated Closed Deals: 12
-----
TOTAL FORECASTED REVENUE: $24000,00
=====
```

NEXT WEEK PREVIEW

WHAT'S COMING

Analytics and dashboards

How is data from ERP and CRM systems transformed into KPIs, reports, and dashboards used by managers to make decisions?